

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:
DOSAGE GUIDELINES

Infection	Type of Severity	Unit Dose	Frequency	Usual Durations #
Acute Sinusitis	Mild / Moderate	500 mg	q 12 h	10 Days
Lower Respiratory Tract	Mild / Moderate Severe/ Complicated	250-500 mg 750 mg	q 12 h q 12 h	7 to 14 Days 7 to 14 Days
Urinary Tract	Acute uncomplicated Cystitis in females (before menopause) Mild / Moderate Severe/ Complicated	125 mg 250 mg 250 mg 500 mg	q 12 h single dose q 12 h q 12 h	3 Days 1 Day 7 to 14 Days 7 to 14 Days
Chronic Bacterial Prostatitis	Mild / Moderate	500 mg	q 12 h	28 Days
Intra-Abdominal*	Complicated	500 mg	q 12 h	7 to 14 Days
Skin and Skin Structure	Mild / Moderate Severe / Complicated	500 mg 750 mg	q 12 h q 12 h	7 to 14 Days 7 to 14 Days
Bone & Joint	Mild / Moderate Severe / Complicated	500 mg 750 mg	q 12 h q 12 h	≥ 4 to 6 wks ≥ 4 to 6 wks
Typhoid Fever	Mild / Moderate	500 mg	q 12 h	10 Days
Gonorrhoea -Extra-genital -Urethral & Cervical	Acute Uncomplicated	125 mg 250 mg	q 12 h single dose	1 Day

* Use in conjunction with metronidazole.

Duration of treatment:

Generally ciprofloxacin should be continued for at least 3 days after the signs and symptoms of infection have disappeared.

Average treatment period: 1 day for acute uncomplicated gonorrhoea and cystitis, up to 7 days for infections of the kidneys, urinary tract and abdominal cavity, throughout the entire neutropenic phase in immunocompromised patients, a maximum of 2 months for osteomyelitis and 7 – 14 days for all other infections.

Treatment should continue for a minimum of 10 days in streptococcal infections.

Infections caused by Chlamydia should be treated for a minimum of 10 days.

Administration

The tablets should be taken whole with a little fluid.

They do not need to be taken at mealtimes. Absorption of the drug is accelerated if taken on an empty stomach.

Elderly patients should receive as low a dose as possible, taking into account the severity of the illness and the creatinine clearance.

Renal and Hepatic Impairment

1.0 Impaired Renal Function:

1.1 Where creatinine clearance is between 31 and 60 ml/min/1.73 m² or where the serum creatinine concentration is between 1.4 and 1.9 mg/100 ml, the maximum daily dose should be 1000 mg/day for oral administration.

1.2 Where creatinine clearance is ≤ 30ml/min/1.73 m² or where the serum creatinine concentration is ≥ 2 mg/100 ml, the maximum daily dose should be 500 mg/day for oral administration.

2.0 Impaired Renal Function + Haemodialysis:

Dose as in 1.2; on dialysis days after dialysis.

3.0 Impaired Renal Function + CAPD:

Administration of ciprofloxacin film coated tablets as 1 x 500 mg film coated tablet or 2 x 250 mg film coated tablets.

4.0 Impaired Liver Function: No dose adjustment is required.

5.0 Impaired Renal and Liver Function: Dose adjustment as in 1.1 and 1.2.

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTES:

Acute overdose

In the event of acute, excessive oral overdose, reversible renal toxicity has been reported in some cases.

Therefore, apart from routine emergency measures, it is recommended to monitor renal function.

Treatment:

Since there is no specific antidote for ciprofloxacin overdose, treatment should be symptomatic and supportive and may consist of the following:

To decrease absorption – Induction of emesis or use of gastric lavage to evacuate the stomach to eliminate unabsorbed drug; Administration of Mg- or Ca-containing antacids.

Specific treatment – Maintenance of adequate hydration.

Only a small amount of ciprofloxacin (<10%) is removed from the body after hemodialysis or peritoneal dialysis.

PACKING / PACK SIZES:

Blister pack of 10's, 3 x 10's/box and 10 x 10's/Box.

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

250 mg: Store below 30°C.

500 mg: Store below 30°C.

Auxiliary labelling.

* May cause dizziness or lightheadedness

* Continue medicine for full time of treatment.

* Avoid too much sun or use of sunlamp.

* Do not take antacids or iron preparations within 4 hours of this medicine.

Shelf life: 3 years

Manufacturer and Product Registration Holder

SM PHARMACEUTICALS SDN BHD (218620-M)

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CIPQUIN (CIPROFLOXACIN) 250 AND 500 TABLET

DESCRIPTION:

250 mg: A white to off-white, round shaped, film coated tablet

500 mg: A white to off-white, capsule shaped, film coated tablet

COMPOSITION:

Each film-coated tablet contains: Ciprofloxacin Hydrochloride equivalent to Ciprofloxacin 250 mg or 500 mg.

ACTIONS AND MODE OR MECHANISMS OF ACTION:

Ciprofloxacin is a DNA gyrase inhibitor used as an antibacterial agent. It is bactericidal and acts intracellularly by inhibiting the A subunit of DNA gyrase (topoisomerase II) which is an essential bacterial enzyme that is a critical catalyst in the duplication, transcription, and repair of bacterial DNA. The process is necessary for compacting the bacterial chromosome within the cell. Other investigators have suggested that the quinolones induce the formation of drug-gyrase-DNA complex through which DNA polymerase cannot proceed. At effective concentrations of ciprofloxacin, selective inhibition of topoisomerase II leads to death of exposed bacteria without producing dysfunction or toxicity of mammalian cells.

Spectrum of activity

Ciprofloxacin has a broader spectrum of activity and is more potent *in vitro* than the non-fluorinated quinolone, nalidixic acid. Activity may be reduced in acid media. Ciprofloxacin is one of the most active of the fluoroquinolones. Among Gram-negative aerobic bacteria, ciprofloxacin is active *in vitro* against Enterobacteriaceae including *Escherichia coli* and *Citrobacter*, *Enterobacter*, *Klebsiella*, *Proteus*, *Providencia*, *Salmonella*, *Serratia*, *Shigella* and *Yersinia* spp. It is also active against *Pseudomonas aeruginosa*, but less so against other *Pseudomonas* spp. *Haemophilus ducreyi*, *H. influenzae*, *Moraxella (Branhamella) catarrhalis*, *Neisseria gonorrhoeae*, and *N. meningitidis* are all very sensitive, including beta-lactamase-producing strains of *H. influenzae*, *M. catarrhalis*, and *N. gonorrhoeae*. Other Gram-negative aerobic bacteria reported to be sensitive to ciprofloxacin have included *Acinetobacter* spp., *Campylobacter* spp., *Gardnerella vaginalis*, *Helicobacter pylori*, *Legionella* spp., *Pasteurella multocida*, and *Vibrio* spp. Variable activity has been reported against *Brucella melitensis*. Among Gram-positive aerobic bacteria, ciprofloxacin is active against staphylococci including penicillinase-producing and -non-producing strains and against some methicillin-resistant strains. Streptococci, in particular *Streptococcus pneumoniae*, and enterococci are less susceptible. Other Gram-positive bacteria sensitive to ciprofloxacin *in vitro* are *Corynebacterium* spp., and *Listeria monocytogenes*. Most anaerobic bacteria, including *Bacteroides fragilis* and *Clostridium difficile*, are resistant to ciprofloxacin. Ciprofloxacin has some activity against mycobacteria, mycoplasmas, rickettsias, and the protozoan *Plasmodium falciparum*. *Chlamydia trachomatis* is not very susceptible and *Nocardia asteroides* and *Ureaplasma urealyticum* are usually considered to be resistant. The spirochaete *Treponema pallidum* and fungi are also resistant.

Activity with other antimicrobial agents

There have been some reports of enhanced activity *in vitro* when ciprofloxacin has been used with other antimicrobial agents such as azlocillin against *Staphylococcus aureus* and *Pseudomonas aeruginosa*, imipenem against *Ps. Aeruginosa*, and cefotaxime or clindamycin against anaerobic bacteria.

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS):

Ciprofloxacin is rapidly and well absorbed from the gastro-intestinal tract. Oral availability is approximately 70% and a peak plasma concentration of about 2.5 µg per mL is achieved 1 to 2 hours after a dose of 500 mg by mouth. The plasma half-life is about 3.5 to 4.5 hours. Plasma protein binding ranges from 20 to 40%.

Ciprofloxacin is widely distributed in the body and tissue penetration is generally good. The volume of distribution is high (approx. 2.8 l/kg). It appears in the cerebrospinal fluid, but concentrations are only about 10% of those in plasma when the meninges are not inflamed. Ciprofloxacin crosses the placenta and is excreted in breast milk. High concentrations are achieved in bile. Ciprofloxacin is eliminated principally by urinary excretion. At least 4 active metabolites have been identified. Oociprofloxacin appears to be the major

urinary metabolite and sulphociprofloxacin the primary faecal metabolite. Urinary excretion is by active tubular secretion as well as glomerular filtration and is reduced by probenecid. Excretion is virtually complete within 24 hours; about 40 to 50% of an oral dose is excreted unchanged in the urine and about 15% as metabolites. Only small amounts of ciprofloxacin are removed by haemodialysis or peritoneal dialysis.

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INDICATIONS:

Uncomplicated and complicated infections caused by ciprofloxacin-sensitive pathogens.

* **Infections of the respiratory tract:** In the treatments of outpatients with pneumonia due to *Pneumococcus*, ciprofloxacin should not be used as a first choice of drug. Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by *Klebsiella*, *Enterobacter*, *Proteus*, *Pseudomonas*, *Haemophilus*, *Branhamella*, *Legionella* and *Staphylococcus*;

* **Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis),** especially if these are caused by Gram-negative organisms including *Pseudomonas* or by *Staphylococcus*.

* **Infections of the eyes**

* **Infections of the kidneys and/or the efferent urinary tract**

* **Infections of the genital organs,** including adnexitis, gonorrhoea, prostatitis.

* **Infections of the abdominal cavity (e.g., infections of the gastrointestinal tract or of the biliary tract, peritonitis).**

* **Infections of the skin and soft tissue.**

* **Infections of the bones and joints.**

* **Septicaemia.**

* **Infections, or imminent risk of infection (prophylaxis) in immunocompromised patients (e.g., those on immunosuppressants, or have neutropenia).**

* **Selective intestinal decontamination in immunosuppressed patients**

Consideration should be given for official guidance on the appropriate use of antibacterial agents.

The following indication(s), should be restricted:

- Acute bacterial rhinosinusitis*
- Acute exacerbation of chronic obstructive pulmonary disease including chronic bronchitis*
- Nosocomial pneumonia / Hospital-acquired pneumonia*
- Acute otitis media*
- External otitis*
- Endocarditis*
- Infection of cerebrospinal fluid*
- Meningitis*
- Septicaemia*
- Uncomplicated acute cystitis / uncomplicated cystitis*
- Prevention of exacerbations in women with recurring urinary tract infections*
- Prevention of infection in surgical procedures in the urogenital system*
- Pre-operative preparations for chronic cholesteatomatous otitis and chronic otitis spreading to bone*

Cipquin should be only used :

- When *Pseudomonas* is considered AND the patient is allergic to antipseudomonal penicillins/cephalosporins;
- For resistant organisms with no other alternative antibiotics available.

Cipquin should not be used >24 hours post operation.

CONTRAINDICATIONS:

Ciprofloxacin should not be given in patients with known hypersensitivity to Ciprofloxacin or other quinolone antibiotics. Ciprofloxacin should not be prescribed to children, growing adolescents and pregnant or nursing women, since ciprofloxacin and related fluoroquinolones have been shown to cause degenerative changes in weight-bearing joints of young animals.

SIDE EFFECTS / ADVERSE REACTIONS:

Effects on the Gastrointestinal Tract: Nausea, diarrhoea, vomiting, digestive disorders, abdominal pain, flatulence, anorexia.

The doctor should be advised of any severe or persistent diarrhoea occurring during or after treatment, since these symptoms could conceal a serious intestinal disorder (pseudomembranous colitis) requiring urgent treatment. In such cases, Cipquin should be discontinued and replaced by another appropriate drug. Preparations which inhibit peristalsis are contraindicated.

Effects on the Nervous Systems: Dizziness, headache, insomnia, agitation, trembling. Very Rarely: Peripheral paralgesia, sweating, unsteady gait, convulsions, anxiety states, nightmares, confusion, depressions, hallucinations. In some instances, these reactions occurred after the first administration of Cipquin already. In these cases Cipquin has to be discontinued and the doctor should be informed immediately.

Hypersensitivity reactions.

Skin reactions, e.g. rashes. Very Rarely: Pruritus, drug fever. Anaphylactic/anaphylactoid reactions (e.g. facial, vascular and laryngeal oedema; dyspnoea progressing to life-threatening shock. In this cases, Cipquin has to be discontinued and medical treatment (e.g., treatment for shock) is required. Punctate skin haemorrhages (petechiae), blister formation with accompanying haemorrhages (haemorrhagic bullae) and small nodules (papules) with crust formation showing vascular involvement (vasculitis), Stevens-Johnson syndrome, interstitial nephritis, hepatitis; very rarely, major liver disorders, including hepatic necrosis.

Effects on the Cardiovascular System: Very rarely: Tachycardia, hot flushes, migraine, fainting.

Effects on Blood and Blood Constituents: Eosinophilia, leucocytopenia, leucocytosis, anaemia; very rarely, thrombocytopenia, thrombocytosis, altered prothrombin levels.

Special senses/sensory organs

Impaired taste and smell, visual disturbances (e.g. double vision, colour vision), tinnitus & transitory impairment of hearing, especially at high frequencies.

Others: Very rarely: Joint pains, general feeling of weakness, muscular pains, tendovaginitis, mild photosensitivity, transient impairment in kidney function, including transient kidney failure.

Effect on Laboratory Values/Urine Deposits: There may be a transient rise in the transaminase and alkaline phosphatase levels, or cholestatic jaundice may occur; transient increase in serum urea, creatinine and bilirubin levels, hyperglycaemia; in individual cases: Crystalluria and haematuria.

Musculoskeletal and connective tissue disorders *Nervous system disorders*

General disorders and administrative site conditions *Psychiatric disorders*

Eye disorders*

Ear and labyrinth disorders*

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

Post Marketing Experience

Exacerbation of myasthenia gravis

Psychiatric disorders

Rare: Depression (potentially culminating in suicidal ideations/ thoughts or suicide

attempts and completed suicide) Very Rare: Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

PRECAUTIONS / WARNINGS:

Hypersensitivity - Cipquin should not be used where there is hypersensitivity to ciprofloxacin or to other chemotherapeutic agents of the quinolone group. In some instances, hypersensitivity and allergic reactions occurred after the first administration; the doctor should be informed immediately. Anaphylactic/anaphylactoid reactions in very rare instances can progress to a life threatening shock. In these cases, Ciprofloxacin should be discontinued and medical treatment (e.g. treatment for shock) is required. Ciprofloxacin should be used with caution in the elderly, in patients with epilepsy or a history of CNS disorders, including cerebral arteriosclerosis (fluoroquinolones may cause CNS stimulation or toxicity).

Musculo-skeletal system - Achilles and other tendon ruptures that required surgical repair or resulted in prolonged disability have been reported predominantly in the elderly on prior treatment with glucocorticoids. Ciprofloxacin should be discontinued, physical exercises be avoided, and a physician be consulted, if the patient experiences pain, inflammation, or rupture of a tendon.

Care is necessary in patients with impaired hepatic or renal function (patients with both hepatic and renal function impairment may require a reduction in the dosage of ciprofloxacin), glucose-6-phosphate dehydrogenase deficiency, or myasthenia gravis. An adequate fluid intake should be maintained during treatment with ciprofloxacin and excessive alkalinity of the urine avoided because of the risk of crystalluria. Exposure to strong sunlight or sunlamps should also be avoided (discontinue if photosensitivity occurs).

WARNINGS

The ability to drive or operate machinery may be impaired by ciprofloxacin, especially when alcohol is also taken.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult physician in an emergency department.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pregnancy: Ciprofloxacin must not be prescribed for pregnant women or nursing mothers since there is no experience on the drug's safety in these patient groups and on the basis of animal studies, it is not entirely improbable that the drug could cause damage to articular cartilage in the immature organism. Animal studies have not shown any evidence of teratogenic effects (malformations). Use of ciprofloxacin is not recommended during pregnancy.

Lactation: Ciprofloxacin is excreted in breast milk. Due to potential risk of articular damage, ciprofloxacin is not recommended during breast-feeding.

The use of Ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with Ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors.

Ciprofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids*. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with Ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

*(For systemically administered levofloxacin-containing products, the listing of risk factors should additionally include: "in patients receiving daily doses of 1000 mg levofloxacin"]

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with Ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Adverse Effects/Undesirable Effects).

Exacerbation of myasthenia gravis Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Psychiatric reactions

Psychiatric reactions may occur even after the first administration of fluoroquinolones, including Cipquin. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops these reactions, Cipquin should be discontinued and appropriate measures instituted. Caution is recommended if Cipquin is to be used in psychotic patients or in patients with a history of psychiatric disease

DRUG INTERACTIONS:

Analgesics - Concurrent administration of fenbufen with quinolone antibiotics may increase the incidence of quinolone CNS adverse effects. Convulsions may occur due to an interaction between the quinolones and NSAIDs.

Iron, Sucralfate or Antacids and highly buffered drugs (e.g. anti-retrovirals) containing Magnesium, Aluminium or Calcium - The absorption of ciprofloxacin is reduced by antacids containing aluminium or magnesium and also by calcium, iron and zinc salts. Consequently, Cipquin should be taken either 1-2 hours before, or at least 4 hours after these preparations. This does not apply to antacids, which do not contain aluminium or magnesium hydroxide (e.g., H₂-receptor blockers). Sucralfate releases aluminium ions in the stomach and thereby reduces the absorption of ciprofloxacin.

Antineoplastics and Immunosuppressants - Absorption of oral ciprofloxacin appears to be reduced after cytotoxic chemotherapy.

Cyclosporin - A transient rise in the concentration of serum creatinine was observed when Cifrobid and cyclosporin were administered simultaneously. Therefore, it is necessary to control the serum creatinine concentrations in these patients frequently (twice a week).

Xanthines - Concurrent administration of ciprofloxacin and theophylline can cause undesirable increase in the serum theophylline concentration. This can lead to theophylline-induced side effects; in very rare cases these side effects can be life threatening or fatal. If concurrent use of the two products is unavoidable, the serum theophylline concentration should be checked and the theophylline dose appropriately reduced.

Warfarin - Concurrent use of warfarin with ciprofloxacin has been reported to increase the anticoagulant effect of warfarin, increasing the chance of bleeding.

Glibenclamide - In particular cases, concurrent administration of ciprofloxacin and glibenclamide can intensify the action of glibenclamide (hypoglycaemia).

Probenecid interferes with renal secretion of ciprofloxacin. Co-administration of probenecid and ciprofloxacin increases the ciprofloxacin serum concentrations.

Metoclopramide accelerates the absorption of ciprofloxacin resulting in a shorter time to reach maximum plasma concentrations. No effect was seen on the bioavailability of ciprofloxacin.